

# SORAIDA QUINTERO

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## SUMMARY

Ph.D. Candidate with strong expertise in statistical analysis, statistical model development, mixed effects and hierarchical modeling, simulation based methods, agent-based modeling, and machine learning techniques used to train, evaluate, and interpret predictive models. Proficient in R and experienced with Python and SQL for data extraction, transformation, exploratory modeling, visualization, and reporting. Skilled in identifying trends, designing and validating analytical models that explain complex human behavior and environmental dynamics, and translating quantitative findings into clear insights for interdisciplinary teams, academic leadership, and non-technical audiences. Proven record of independent research, project leadership, and cross-disciplinary collaboration, with particular interest in applying rigorous data driven and model based approaches to decision making, performance evaluation, and institutional improvement.

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## EDUCATION

**Ph.D. Candidate**, University of Illinois Chicago

*Fall 2026*

**M.S.**, Purdue University

*Dec 2021*

**B.S.**, Chicago State University

*May 2016*

### Relevant Coursework:

Biostatistics, Machine Learning, Individual-based modeling, Statistical Methods Biology, Quantitative Methods, R programming, Stats for Industry, Stats for Natural Resources, SQL

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## TECHNICAL SKILLS

### Programming & Data:

R (tidyverse, dplyr, purrr, ggplot2, vegan); Python (pytorch, pandas, numpy, NLP/text extraction workflows); SQL; Bash scripting; Linux/Unix environments; GitHub; Jupyter Notebook, AWS

### Statistical Modeling & Methods:

Generalized linear models, logistic regression, mixed effects model interpretation, clustering and unsupervised learning, agent-based modeling, spatial statistics, simulation modeling (Monte Carlo techniques, parallel)

**Machine Learning & Predictive Analytics:**

Supervised and unsupervised learning; regression and classification models; XGBoost, gradient boosting, decision trees, random forest; model comparison; cross-validation; feature engineering; topic modeling; co-occurrence network analysis; NLP pipelines; image identification

**Data Processing & Analysis:**

Data wrangling, reproducible workflows, exploratory data analysis, text mining, structured and unstructured data integration, real-world/clinical-data readiness (cleaning, validation, preprocessing)

**Languages:**

Spanish (fluent)

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**RESEARCH EXPERIENCE****Graduate Research Assistant** — University of Illinois, Chicago

*2021 – Present*

- Designed and validated agent-based and statistical models to evaluate how individual behaviors scale up to city wide level outcomes.
- Built generalized linear and logistic regression models to quantify how environmental and social factors influence spatial patterns.
- Conducted spatial statistical analysis (Moran's I, Mantel r) to identify clustering and decision mimicry.
- Performed regression and mixed-model analysis to evaluate socio-ecological responses to environmental change.
- Communicated analytical insights to interdisciplinary researchers, community stakeholders, and scientific audiences.

**Graduate Assistant** — Purdue University, West Lafayette

*2016 – 2021*

- Managed project coordination and day-to-day operations for an interdisciplinary Mellon Foundation grant, including meeting facilitation, milestone tracking, and communication across faculty and administrative units.
- Developed stakeholder outreach plans and communication strategies to translate research findings into accessible formats for university leadership, policymakers, and external partners.
- Prepared formal reports, data summaries, and budget documentation to support institutional accountability and ensure compliance with foundation reporting requirements.
- Planned and coordinated events, workshops, and presentations to support interdisciplinary collaboration and increase visibility of research activities across campus.
- Collaborated with policy experts, communications teams, and Purdue University Press to produce high-quality research outputs and support evidence-based decision-making within the institution.

## TEACHING EXPERIENCE

**Teaching Assistant, BIOS480: Biostatistics with R** — University of Illinois, Chicago  
*2021 & 2023 – Present*

- Assisted undergrad and graduate students in applying statistical tools such as regression modeling, ANOVA, non-parametric tests and model diagnostics.
- Guided students in applying statistical reasoning to various datasets and practiced problem solving techniques.
- Mentored students in experimental design and quantitative reasoning throughout course projects.

**Lab Assistant, BIOS312: Cell Biology Lab** – University of Illinois, Chicago  
*2022*

- Prepared laboratory materials, solutions, reagents, and equipment for weekly cell biology experiments, ensuring accuracy, sterility, and proper storage conditions.
- Set up and maintained laboratory workstations, microscopes, and safety apparatus prior to each class, supporting smooth and efficient lab operations.
- Managed inventory of consumables and chemicals, restocked supplies, and adhered to safety and waste-disposal protocols to maintain a compliant and organized teaching laboratory.

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## SELECTED PROJECTS

### **Spatial Mimicry in Chicago Residential Landscapes**

*Examined how social interactions and neighborhood factors shape spatial patterns of residential yard management.*

**Methods:** Applied spatial statistics and logistic regression to assess plant clustering patterns by measuring spatial autocorrelation using Moran's I and Mantel r tests.

**Outcome:** Demonstrated that social behaviors at the neighborhood level are spatially autocorrelated, revealing mimicry patterns among adjacent households.

### **Neighborhood Gentrification Predictions through Real Estate Market Analysis**

*Created a property analysis model to predict neighborhood gentrification patterns using property and area characteristics with publicly available housing and census data.*

**Methods:** Regression and clustering in R using `tidyverse`, `ggplot2`, and machine learning algorithms (random forest, gradient boosting).

**Outcome:** (preliminary) Improved property value prediction accuracy and identified socio-economic indicators driving neighborhood change.

### **Free-Roaming Dog Population Modeling**

*Developed agent-based models to simulate dog population dynamics, and feeding behaviors across diverse ecological and cultural contexts.*

**Methods:** Implemented demographic and movement modeling, intervention simulations (vaccination, sterilization), and spatial analyses

**Outcome:** (preliminary) Identified context-specific drivers of population level change, and demonstrated that targeted interventions for free-roaming dogs yield significantly improved management outcomes.

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## **PUBLICATIONS**

Quintero S, Ballut N, Johnson P, Minor E, (2025). *Examining the mechanisms that create spatial patterns in vegetation across residential landscapes*. Computers Environment and Urban Systems. Manuscript submitted.

Quintero S, Zollner P, Atallah S, (2025). *Understanding the Disturbance of Human Recreation on Wildlife using Multiple Dynamic Agents within an IBM Framework*. Ecological Modelling. Manuscript in progress.

Quintero S, Weldon L, Clawson R, Roberts A, (2024). *Engaging Communication for High Impact Research: Four models for organizing interdisciplinary research teams*. Proceedings of the National Academy of Sciences. Manuscript in progress.

Quintero S, Minor E, Railsback S, Johnson P, (2023). *Yards* (Version 1.0.0). CoMSES Computational Model Library.

Quintero S, (2022). *Making the change*. The Wildlife Professional, 16(2), 49–51.

Hernandez G, Quintero S, Vilela J. F., & de La Sancha N. U, (2017). *Ontogenetic variation of an omnivorous generalist rodent: The case of the montane akodont (Akodon montensis)*. Journal of Mammalogy.

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## **CONFERENCE PRESENTATIONS & POSTERS**

Quintero S, Minor E. Neighbor Mimicry and Plant Nurseries as Drivers of Spatial Patterns in Urban Residential Landscapes. IALE Conference; 2023 March 19; Riverside, California.

Quintero S, Cohen A, Atallah S, Zollner P. Understanding the Disturbance of Human Recreation on Wildlife Using Multiple Dynamic Agents Within an IBM Framework. BTAA GIS Conference; 2020 Nov 13; Virtual

Quintero S, Cohen A , Atallah S, Zollner P. Understanding the Disturbance of Human Recreation on Wildlife Using Multiple Dynamic Agents Within an IBM Framework. Poster presentation. The Wildlife Society Conference; 2020 Sept 28 – Oct 2; Virtual

Quintero S, Weldon L, Clawson R, Roberts A. Engagement models of science communication: *Working tools for team researchers at a micro level*. Poster presentation at: International Network for the Science of Team Science. SciTS Conference; 2019 May 20-23; Lansing, MI. ***Best poster award***

Quintero S, Kirt S. Ant's in the Prairie: A comparison of Midewin National Tallgrass Prairie's Remnant and Restored Areas. Poster presentation at: LSAMP International Undergraduate Research Conference, 2014; Tarlac, Philippines

Quintero S, Kirt S. Ant's in the Prairie: A comparison of Midewin National Tallgrass Prairie's Remnant and Restored Areas. Oral presentation at: Argonne National Laboratory, 2013; Darien, IL

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## **WORKSHOPS, FELLOWSHIPS & AWARDS**

Andrew Mellon Fellow • Alliance for Graduate Education Scholar • ESA Seeds Grant • NCEAS Director's Scholar • DaSEH, R Workshop, Fred Hutch Cancer Center

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## **PROFESSIONAL MEMBERSHIPS**

Ecological Society of America • American Society of Mammalogists • Pycon • Bio Grad Assoc

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## **OUTREACH & ENGAGEMENT**

Mujeres Latinas en Acción (Battered Women's Shelter) • Great Lakes Coastal Restoration • Prairie Conservation & Community Education • Biology Graduate Student Association

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## **REFERENCES**

**Dr. Emily Minor**  
Professor, University of Illinois Chicago  
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**Dr. Pat Zollner**

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**Dr. Gaby Nunez-Mir**

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